

The Impact of Tuition Increases on Enrollment at Public Colleges and Universities

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In this paper we review recent increases in tuition at public institutions and estimate impacts on enrollment. We use data on all U.S. public 4-year colleges and universities from 1991 to 2006 and illustrate that tuition increased dramatically beginning in the early part of this decade. We examine impacts of such increases on total enrollment and credit hours, and estimate differences by type of institution. We estimate that the average tuition and fee elasticity of total headcount is -0.0958. At the mean, a \$100 increase in tuition and fees would lead to a decline in enrollment of about 0.25 percent, with larger effects at Research I universities. We find limited evidence that especially large tuition increases elicit disproportionate enrollment responses.

Keywords: tuition, costs, postsecondary enrollment, public higher education

In the summer of 2009, with its budget a shambles, the state of California cut support for the University of California system by 20%, or over \$800 million (O’Leary, 2009). The allocation to the California State University system suffered a similar cut of 20% (Gordon, Holland, & Landsberg, 2009). Faced with this sharp decline, the University of California Board of Regents approved a tuition and fee increase of more than 32%, setting off a storm of protests on campuses throughout the system (Lewin & Cathcart, 2009). Although California’s situation is severe, it is hardly unique. With economic conditions weak and financial pressures on state budgets growing, the *New York Times* reported that the

need to offset these declines by raising college tuition was building rapidly (Lewin, 2008). This upward pressure on tuition in public higher education is not new, and neither is the declining general revenue support by state government (Johnstone, 2004; Koshal & Koshal, 2000; Rizzo & Ehrenberg, 2003). In the case of California, in 1990, the state appropriation to the University of California system amounted to \$16,430 per student, compared with \$7,570 in the most recent year (Friend, 2010, p. 24).

As fiscal pressures have mounted, college and university administrators and their governing boards have been forced to offset declines in nontuition sources of revenue. Naturally, they face

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substantial pressure to increase tuition. The faculty representative to the University of California regents characterized this pressure as follows: “The legislators have told us, essentially, ‘The student is your A.T.M. They’re how you should balance the budget’” (Friend, 2010, p. 24). Although administrators and analysts are aware, at least at some level, that demand schedules are downward sloping, the implicit assumption among many higher education administrators seems to be that tuition elasticity of enrollment is tolerably small, so that any enrollment decline will be small enough that net revenues will rise with the higher tuition. More generally, an important concern is whether rising prices are making higher education less affordable.

In this article, we examine the impact on enrollment of rising tuition at public colleges and universities. Our first objective is to update estimates of the price sensitivity of enrollment in public education. Much of what we know about enrollment response to tuition increases comes from data collected in the 1980s, an era predating the recent run-up in tuition and fees. Using data from the Integrated Postsecondary Education Data System (IPEDS) on all 4-year public institutions from 1991 to 2006, we estimate the relationship between tuition increases and enrollment. We examine the impact of price changes on several measures of enrollment: total headcount, total number of credits taken, and the number of first-time, full-time (FTFT) freshman. As we describe below, the first of these measures is likely to be the slowest to respond but is clearly an important measure of demand for any institution.

The data we use constitute an institution-level panel and permit us to measure enrollment, tuition and fees, and student aid for all public 4-year colleges and universities. In addition, we can measure the size of the cohort of high school graduates within a state and the tuition and fees charged at all other 2- and 4-year colleges within the state. The identifying assumptions for our models are that tuition increases are largely exogenous (at least at the institution level) and that schools have limited power to affect enrollment by modifying admissions decisions when adopting tuition increases. As we discuss in this article, both of these assumptions appear to hold. For example, a large portion of the within-state variation in tuition and fees is common across institutions,

and we find no evidence that an institution’s admissions decisions are related to tuition and fee changes.

A second objective derives from an important aspect of intertemporal patterns of tuition at public institutions. Although tuition levels can vary substantially across institutions within a state, as we illustrate below, tuition patterns over time are quite similar across institutions within states. Although tuition may be rising in similar ways across various public institutions, the markets they serve can be quite dissimilar. Public institutions are diverse, with different missions and students. As an example, though in the same state, the University of Michigan in Ann Arbor and Lake Superior State University differ along many dimensions important for understanding comparative statics of price changes.¹ Because of variety in the education and experience colleges and universities provide students, and because of differences in the financial resources of the average student across institutions, enrollment responses to price fluctuations may vary. So, one of our goals is to assess the degree to which the impact of tuition increases on enrollment varies at different types of institutions.

Finally, we examine the impacts of exceptionally large tuition increases on enrollment. Recent declines in nontuition revenue have forced administrators at public colleges and universities to adopt unusually large tuition increases. To give some sense of the magnitude of the tuition increases recently implemented across the country, during the early part of the decade, as states grappled with fiscal crises, the University of Arizona increased real tuition from \$2,906 to \$3,948 for a full-time in-state student in 2003. The same year, the University of Massachusetts–Lowell increased tuition from \$5,842 to \$8,040. Changes such as these not only raise public awareness of the rising cost of higher education but also make more important the task of updating estimates of enrollment impacts. One question we address is whether large, abrupt changes in tuition such as these have disproportionately large effects on enrollment.

In the next section, we briefly summarize recent work on the relationship between tuition costs and enrollment in higher education in the United States and summarize recent trends in tuition. We then describe our empirical objectives

and estimation strategy, along with the IPEDS data we use. We then turn to our examination of patterns of tuition and enrollment at 4-year public colleges and universities during the past two decades. We not only focus on average impacts and basic revenue implications but also explore the degree to which student response differs across types of public institutions. We also examine enrollment changes following exceptionally large tuition increases that have typified price changes in the most recent decade. Finally, we consider evidence relevant to understanding what information these enrollment changes provide about the tuition elasticity for demand.

Background

Price Response Literature in Higher Education

Economists and other analysts have long been interested in understanding the demand for higher education. Examples of such work include studies focused on quantifying price elasticities for various student populations, estimating student sensitivity to changes in financial aid packages, or constructing university-specific demand functions.

Much of the early work on the demand for higher education was reviewed by Jackson and Weathersby (1975). Using the parameters estimated in a number of studies, they concluded that the net behavioral response to changes in tuition is modest: a decrease of between 0.05% and 1.46% in enrollment ratio per each \$100 increase (in 1974 dollars) in student cost. Additionally, they found the absolute magnitude of price responsiveness to decrease with increasing income. In a meta-analysis of studies completed between 1967 and 1982, Leslie and Brinkman (1987) concluded a \$100 tuition price (in 1982 dollars) increase to be associated with a 0.6 to 0.8 percentage point decline in college enrollments.

Heller (1997) provided an update to Leslie and Brinkman (1987). He concluded that a \$100 increase results in a 0.5% to 1.0% decline in enrollments. But he pointed out that the empirical work he examined used data from the 1970s and 1980s, so the effect might not generalize to the higher tuition levels at the time of his analysis (p. 650). A decade after Heller's analysis, tuition has climbed higher still.

Although much of the initial work on enrollment responses to tuition consisted of institutional demand studies, several studies examined national-level data. For example, Kane (1994a) and St. John (1990) used the High School and Beyond data (which followed a sample of students who were high school sophomores and seniors in 1980) and estimated that enrollments fell by approximately 0.5% to 1% with a \$100 increase in tuition. Rouse (1994) used data from the National Longitudinal Survey of Youth, a cohort contemporary to the High School and Beyond students, and found that an 8% increase in tuition resulted in a decline in enrollments of between 0.67% and 1%. Using pooled cross-sections of the October Current Population Survey from 1973 to 1988, Kane (1994b) found a \$1,000 increase in the net direct costs of college to be associated with a 5 percentage point decline in the likelihood of attending college. Most relevant to our empirical work, Heller (1996) and Kane (1995) used data from the IPEDS for the 1980s and early 1990s. In both cases, they found that a \$100 increase in tuition at 4-year institutions resulted in a decline in enrollments of just under 0.5%.

Many other studies have exploited changes in financial aid packages or laws to estimate the enrollment impact of the costs of higher education. For example, Kane (2003) examined the Cal Grant aid program in California using data from 1998 and 1999. He found that applicants to the financial aid program were 3 to 4 percentage points more likely to enroll in college. Dynarski also examined evidence from both the HOPE scholarship program in Georgia (Dynarski, 2000) and the elimination of the Social Security Student Benefit Program in 1982 (Dynarski, 2003). In both studies, she found each \$1,000 increase in aid to be associated with an increase in the college attendance rate of about 4 percentage points.²

One recent study examined enrollment effects specifically at public universities and colleges. Shin and Milton (2006) used IPEDS data on FTFT fall enrollments and in-state tuition levels from 1998 to 2002 to estimate the impacts of tuition increases on enrollment growth. They concluded that enrollment changes were not affected by changes in tuition or financial aid over this time period (p. 234). Yet because of missing outcome data, Shin and Milton could analyze only

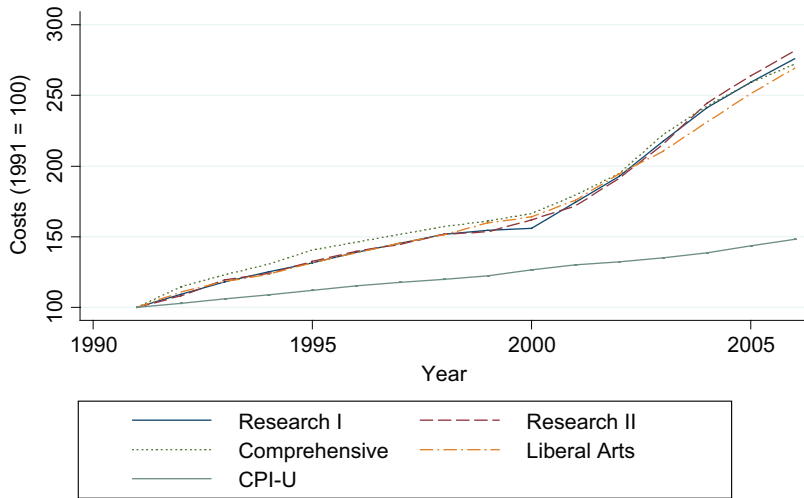


FIGURE 1. *Tuition costs at public universities and colleges.*

Source: Data for the Consumer Price Index for all urban consumers (CPI-U) are from the Bureau of Labor Statistics.
Note: Tuition prices are enrollment weighted.

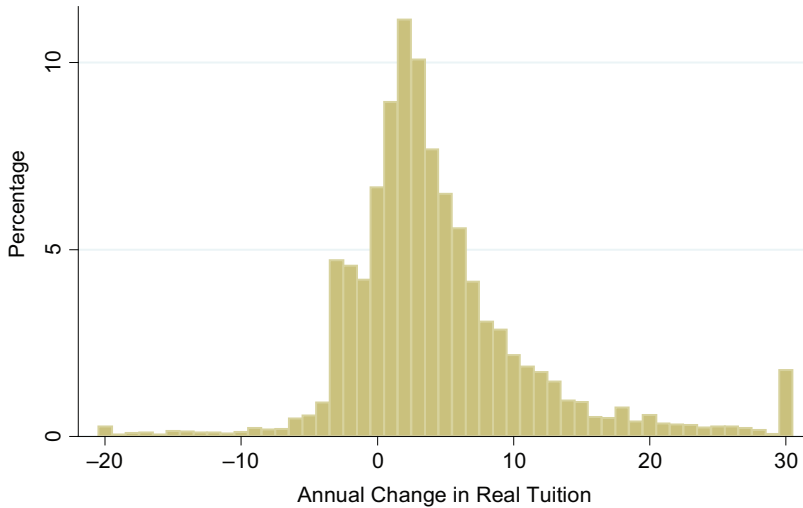


FIGURE 2. *Distribution of year-to-year changes in real tuition, 1991 to 2006.*

3 years of data. This leaves little identifying information on changes in enrollments and tuition costs within institutions over time. Furthermore, some of the largest average year-to-year changes in tuition price occurred after the end of Shin and Milton's (2006) panel.

Recent Trends in Tuition Costs at Public Colleges and Universities

With the exception of Shin and Milton's (2006) analysis, these previous studies have all used data

from about a decade or more ago. In the meantime, tuition has continued to rise, and by a lot. Using data from IPEDS and the U.S. Bureau of Labor Statistics, Figure 1 shows the increase in nominal tuition costs from 1991 to 2006 compared with the increase in the inflation rate, as measured by the Consumer Price Index (CPI).³ Clearly, tuition costs are rising substantially, and the rate of increase accelerated recently. Additionally, tuition rose at comparable rates at research-intensive universities, comprehensive universities, and liberal arts colleges.

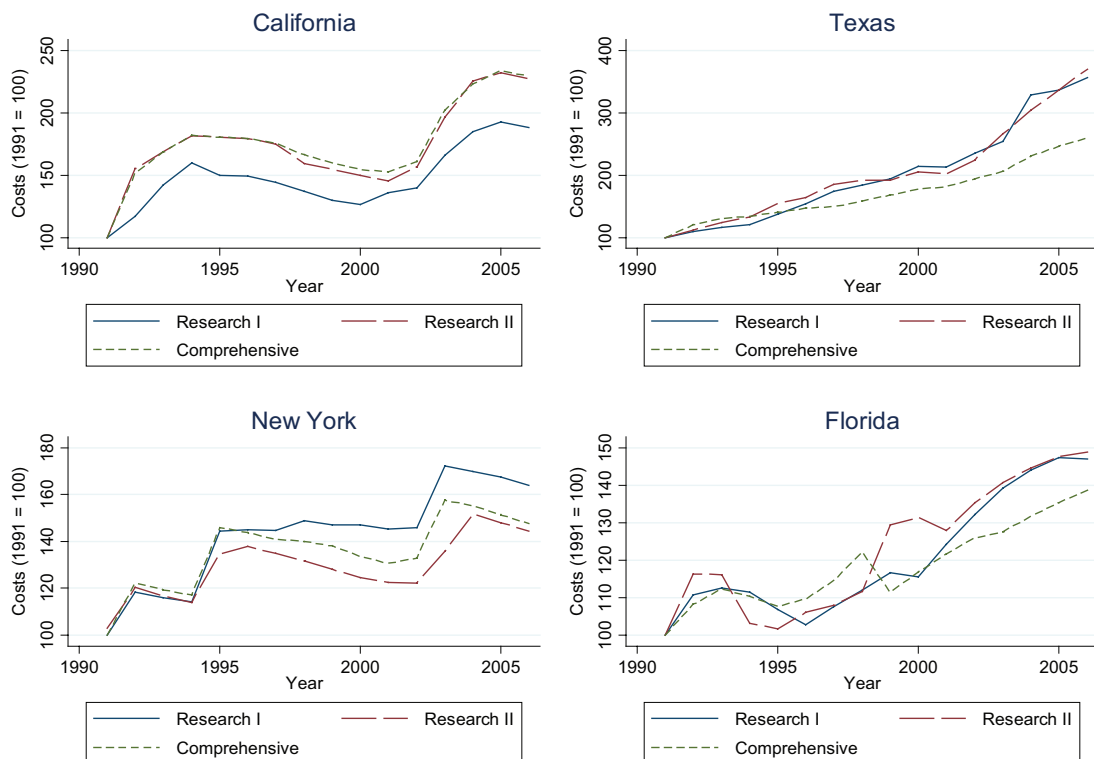


FIGURE 3. Trends in tuition costs in selected states, by institution type, 1991 to 2006.

The time series of average tuition in Figure 1 does not illustrate two important stylized facts about recent changes in tuition relevant to our work. First, the distribution of year-to-year real tuition increases is positively skewed. The mean annual increase in real tuition during the period we study was 4.2%. But a number of institutions implemented much larger real year-to-year increases. In Figure 2, we present the frequency distribution of real year-to-year tuition increases for all public 4-year colleges and universities from 1991–1992 to 2006–2007. Although many of these year-to-year increases are in the neighborhood of 4% or 5%, a considerable number are above 10%, 15%, and even 20%.

The second important fact about recent changes in public higher education costs is that although there are markedly different intertemporal changes in tuition across states, within states, trends in tuition are generally similar across the various types of 4-year institutions. So, although Figure 1 illustrates that at the national level, patterns of average changes in tuition are similar for various

types of institutions, we find the same thing within states. We illustrate this in Figure 3, which shows the time series of real tuition (indexed to 1991) at public Research I, Research II, and comprehensive universities in the four most populous states in the United States. Generally, tuition increases across public 4-year institutions are very similar within states. This is to be expected, because states typically govern public colleges and universities through multicampus systems and have established higher education coordinating boards for the purposes of advancing general policy objectives across campuses. On average, we find that about 68% of the within-state variation in tuition costs is common across institutions.⁴

Identifying the Elasticity of Demand

Our objective is to understand how these recent tuition increases affect demand for education at public 4-year colleges and universities. As with any good or service, price increases such

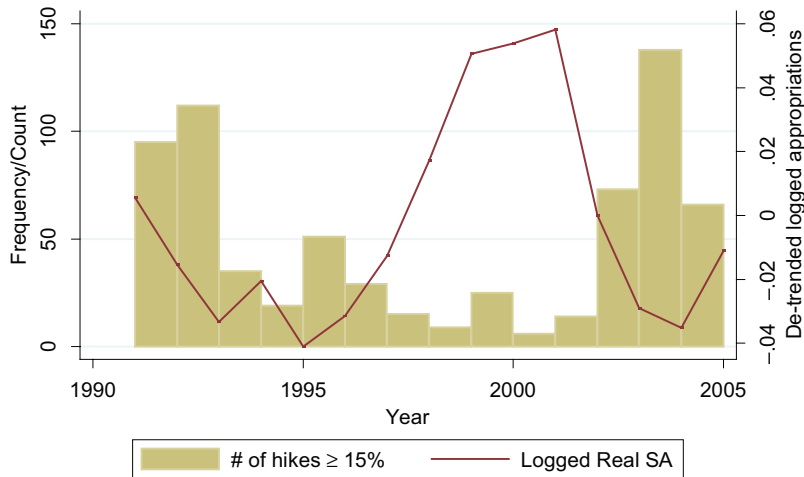


FIGURE 4. *Temporal distribution of large tuition hikes versus logged real state appropriations (SA).*
 Note: State appropriations are available only through the 2005–2006 academic year.

as these could arise because supply has shifted due to suppliers leaving the market. But we find no evidence of this: The number of public 4-year institutions has increased modestly over the past decade, from 622 in 1997 to 638 in 2006 (National Center for Education Statistics, 2008). Contemporaneous to this modest change was a large change in total enrollment: Between 1995 and 2005, total enrollment at public 4-year postsecondary institutions grew by slightly more than 1 million students (National Center for Education Statistics, 2005a). In any case, because we focus on 4-year public institutions, it is important to recognize that the processes that set prices here do not necessarily adhere to market principles. In previous studies, researchers have found that much of the within-institution variation in tuition at public universities appears to be driven by fluctuation in state appropriations (Koshal & Koshal, 2000; Rizzo & Ehrenberg, 2003). Furthermore, Lowry (2001a, 2001b) presented empirical evidence that factors determining fluctuation in state appropriations are exogenous to the process shaping enrollment demand.

Evidence from the IPEDS, too, suggests that state appropriations drive pricing decisions at 4-year institutions. As an illustration, consider again the large tuition increases in the upper tail of Figure 2. In Figure 4, we present a frequency distribution of the number of tuition hikes of 15% or more at public 4-year institutions over the course of our panel. Superimposed on this dis-

tribution is a time series of the mean of detrended logged real state appropriations received by these institutions in each year, net of institution fixed effects. If the average institution's revenue growth followed the national trend, this line would be flat, at zero. In years when institutions receive real appropriations in excess of their normal appropriations (detrended), the series is positive. In bad years (lower appropriations than typical), the series is negative. Clearly, real state appropriations were about 4% below typical levels in the early part of this decade, when most of the large year-to-year tuition hikes occurred. During the late 1990s, when appropriations were relatively high, there were few large tuition increases. Our reading of these patterns is that large tuition increases at public colleges and universities track changes in state appropriations.

One interpretation of the substantial relationship between state appropriations and tuition is that at the institution level, administrators are highly constrained in their ability to set tuition. As such, a good bit of the variation in an institution's tuition can be treated as exogenous, so subsequent relationships between tuition and enrollment can provide information about demand elasticity. In practice, however, the collective research on the tuition price and enrollment relationship has done little to model the simultaneous relationship between the two.⁵ Rather, researchers have focused on using variation in tuition prices, conditioning on demographic controls such as the number of high school graduates, as

well as institution effects and year controls. We adopt a similar strategy here to limit the possibility that the observed tuition enrollment relationship is due to shifts in supply of potential college students.

There is some evidence on shifts in the aggregate supply of college students to the labor market over long periods. For example, Bound and Turner (2006) provided evidence that over the past half century, large cohorts of college-age students faced higher net costs and subsequently lower rates of undergraduate degree attainment. We focus here on a much shorter period, for which these demographic shifts can have a limited role, if any. Within our shorter period, if states where costs of higher education were rising faster also saw above-average growth in the size of the college-age population, this could lead us to understate the impact of price changes on enrollment. On the other hand, if these states saw the smallest changes in the number of young persons, we might overstate the demand response to price increase. To explore the direction of any potential bias in our own data, we calculated mean changes in the size of the graduating high school cohorts for states and years in the top and bottom quartiles of the distribution of year-to-year tuition changes. We found no statistically significant difference in the growth of the number of high school graduates between states with the fastest growing tuition costs and those with the slowest growing costs. Furthermore, to limit these concerns, all our empirical models include year fixed effects. To the extent that there were important shifts in the number of persons of college age, or applying for college, within our panel, these year effects will pick up any common effects on enrollment demand. More important for estimating demand elasticity are matters of model specification that control for quality differences, income, labor market conditions, and prices of substitutes.

Methods

To investigate student response to tuition increases at U.S. public universities, we use IPEDS data from 1991–1992 to 2006–2007. The IPEDS is an institution-level data set, and we use data on all reporting 4-year public institutions across the entire United States. We focus on 4-year public universities because the vast majority of students

at 4-year colleges and universities attend public institutions. In 2003–2004, public universities enrolled 69% of all undergraduate students enrolled at 4-year institutions (National Center for Education Statistics, 2005b). Furthermore, as a matter of public policy, demand functions at public institutions are of primary concern.

IPEDS is the main postsecondary education collection program from the National Center for Education Statistics. It is a system of survey components designed to collect data from all organizations whose primary purpose is to provide postsecondary education. IPEDS contains a compilation of institution-level data on enrollment, program completion, faculty size and salaries, staff, institutional prices, and other institutional characteristics (National Center for Education Statistics, n.d.). IPEDS provides three different yearly measures of enrollment: total undergraduate unduplicated headcount, total undergraduate credit hours, and the total number of FTFT undergraduates in the entering cohort. Although the first two measures are available for the majority of our panel, the FTFT measure is available only from 1999–2000 to 2006–2007.

The Bureau of Labor Statistics reports on many key national and regional economic indicators. We use yearly state unemployment rate statistics from the Bureau of Labor Statistics, personal income measures by state from the U.S. Bureau of Economic Analysis, and data on state populations from the U.S. Census Bureau. We also use data on the number of public high school graduates by state and year from the *Digest of Education Statistics*, published by the National Center for Education Statistics. For clarity, we provide additional information in the Appendix on the specific source, name, and availability of each of our main variables.

For our panel, we estimate log-log models of the following general type:

$$\begin{aligned} \ln(EN_{it}) = & a + b_1 \ln(T_{it}) + b_A \ln(Aid_{it}) \\ & + b_{HS} \ln(HSgrads_{st}) + b_{CP} \ln(CP_{st}) \\ & + b_I \ln(Inc_{st}) + b_U \ln(Un_{st}) + a_i + a_t + e_{it}. \end{aligned} \quad (1)$$

We use log-log specifications for ease of interpretation and because of improved model fit, where EN_{it} is a measure of enrollment at institution i in academic year t . We use the three different measures of enrollment described above. Because students can adjust credit hours and freshman

enrollment more readily at the margin, we anticipate total headcount to be the least responsive to price changes.⁶

The key independent variable is T_{it} , a measure of in-state tuition and fees charged to full-time students attending institution i in year t .⁷ Aid_{it} is a vector of two measures: total Pell Grant dollars disbursed or otherwise made available to recipients by the institution and the total gross amount of scholarships and fellowships awarded. Student aid is an important mechanism for reducing the real price of postsecondary education. Among full-time students attending 4-year public institutions, about 76% reported receiving some kind of financial aid during the 2003–2004 academic year (National Center for Education Statistics, 2006).⁸ Furthermore, the proportion of full-time students using institutional aid increased from 17% in 1992–1993 to about 23% in 1999–2000. Over this same period, the average amount of aid received by these students increased (in constant 1999 dollars) from \$2,200 to \$2,700 (Horn & Peter, 2003). Indeed, public institutions faced with implementing large tuition increases might attempt to offset those costs by making larger financial aid offers or better facilitating the ability of students to apply for federal aid or subsidized loans (Marklein, 2002).⁹ We include a number of controls to capture institution- and state-level factors that affect enrollment for a given institution and over time. $HSgrads_{st}$ is a measure of the number of public high school graduates in state s in academic year t . CP_{st} is a vector of competitors' prices, including average community college tuition and fees and average 4-year private university tuition and fees, in state s in year t . Inc_{st} and U_{st} are measures of average per capita income and the unemployment rate for state s in year t , respectively. We include these to control for economic and social conditions that could potentially affect both enrollment numbers as well as the opportunity costs students face.

We include institution-specific fixed effects (α_i) and year effects (α_t). By including institution fixed effects, we use within-institution variation over time in tuition and fees to estimate enrollment effects, net of common year effects. To account for the possibility of serial correlation in the error term, we cluster standard errors at the institution level.¹⁰

To this basic setup, we carry out two main extensions. First, we estimate this general model separately by Carnegie classification of institution (Carnegie Foundation for the Advancement of Teaching, 2006). The strategy to identify effects of tuition changes on enrollment described above will provide the average within-institution effect for a group of heterogeneous institutions: large research-intensive and doctoral-granting universities, comprehensive teaching universities, and small liberal arts colleges. Large research-intensive universities are classified by Carnegie as Research I or Research II institutions. Research I universities include the University of Wisconsin–Madison, Florida State University, and Stony Brook University. Research II universities include Wichita State University, East Carolina University, and The College of William & Mary. Comprehensive institutions grant fewer doctorate degrees but at least 50 master's degrees. Liberal arts colleges primarily grant undergraduate degrees. For example, the University of Michigan–Flint, San Jose State University, and Montclair State University are comprehensive universities, while Evergreen State College, St. Mary's College of Maryland, and the University of Minnesota–Morris are liberal arts colleges.

Not only do these institutions vary in size and mission, but their students may vary in their sensitivity to price changes. Furthermore, the importance of tuition as a source of revenue varies by institution type (National Center for Education Statistics, 2005c). Hence, the ability to provide learning, campus activities, or services to students may be more sensitive to fluctuations in tuition revenue at comprehensive universities and liberal arts colleges than at research-intensive universities.

Second, we examine whether enrollment following especially large year-to-year tuition hikes falls, over and above the enrollment response resulting from the tuition increase itself. Especially large tuition increases from one year to the next might elicit a response from students, over and above a scaled-up response to a more modest tuition increase. Students may view large tuition hikes as unfair, out of line with previously established norms, or even violating an implicit contract once enrolled. Large tuition hikes may also signal that a school is performing and planning

TABLE 1
Descriptive Statistics

Variable	<i>n</i>	<i>Mean</i>	<i>SD</i>
Costs			
In-state tuition and fees	7,075	4,209.69	1,668.29
Out-of-state tuition and fees	7,029	10,845.35	3,622.05
Average community college tuition and fees	7,075	2,036.19	852.14
Average 4-year private college tuition and fees	7,075	16,507.81	4,241.29
Total scholarships and fellowships (× \$1,000)	7,075	20,500.00	25,500.00
Total Pell Grants (× \$1,000)	7,075	6,319.30	5,222.99
Outcomes			
12-month unduplicated headcount (enrollment)	7,075	10,714.98	8,496.81
Total undergraduate credit hours	7,075	261,076	342,050
Total FTFT freshman enrollment (cohort)	3,831	1,556.67	1,357.22
FTFT freshman enrollment (in state)	3,831	1,305.20	1,129.67
FTFT freshman enrollment (out of state)	3,831	251.45	386.54
Percentage FTFT in state	3,831	82.07	17.36
Percentage FTFT out of state	3,831	15.55	13.83
Other controls			
Average per capita income, by state	7,075	32,312.83	4,888.09
Annual state unemployment rate	7,075	5.13	1.25
State population	7,075	9,276,186	8,463,302
Number of high school graduates, by state	7,075	84,642.65	79,162.42

Note: FTFT = first-time, full-time. All monetary values are expressed in 2006 dollars. The analytical sample includes a total of 557 institutions.

poorly or is in dire financial straits, or they may act as an indication of price changes to come. Or very large year-to-year tuition increases may have disproportionately large enrollment effects because families planning and saving for college may not be able to adjust long-term financial planning to accommodate abrupt price changes.¹¹

To examine if large tuition increases have proportionately more impact on enrollment than small hikes, we modify the specification of our empirical model. In the augmented model, described in Equation 2, in addition to the contemporaneous direct measure of tuition and fees, we include a separate vector of dummies defined around a large hike in tuition at institution i ($Hike_{it}$) between year $t - 1$ and year t . If large hikes have proportionately no more impact on enrollment than smaller tuition increases, b_T will capture the entire impact of the tuition increase. But if large hikes elicit an enrollment response more than proportionate to their size, we expect $b_H < 0$.

$$\ln(EN_{it}) = a + b_T \ln(T_{it}) + b_H Hike_{it} + b_A \ln(Aid_{it}) + b_{HS} \ln(HSgrads_{it}) + b_{CP} \ln(CP_{it}) + b_I \ln(Inc_{it}) + b_U \ln(Un_{it}) + a_i + a_t + e_{it} \quad (2)$$

We explore various thresholds for determining what constitutes a large hike. But the basic idea is that we include an indicator equal to 1 if institution i increased tuition between year $t - 1$ and year t in excess of some threshold (e.g., 10%, 15%, or 20%). Because students often enroll at an institution as freshmen with the intention of graduating from that institution, those already enrolled at the time of large increases may be less responsive to tuition hikes than prospective students. To see if the full effect of large tuition hikes develops over a few years, we include lags of the main hike dummy variables to capture the enrollment effect in years following large tuition hikes.

Results

Descriptive Statistics

We begin by considering descriptive statistics on enrollments, tuition rates, school-level characteristics, and state-level characteristics for our sample, presented in Table 1. Over the course of the panel, the average annual cost of in-state

TABLE 2
Enrollment Effects of Tuition Increases: Full Sample

Variable	Log headcount	Log credit hours	Log FTFT
Log tuition and fees (in state)	−0.0958 (0.0312)***	−0.1089 (0.0339)***	−0.1147 (0.0964)
Log average private 4-year tuition and fees	0.0091 (0.0326)	0.0335 (0.0437)	0.1376 (0.1162)
Log average community college tuition and fees	0.0093 (0.0280)	0.0002 (0.0342)	0.0391 (0.0676)
Log average per capita income	0.3223 (0.1640)**	0.2164 (0.1849)	0.0470 (0.4599)
Log number of high school graduates	0.3221 (0.0664)***	0.4096 (0.0684)***	0.5740 (0.1473)***
Log unemployment rate	0.0015 (0.0248)	−0.0164 (0.0294)	0.0547 (0.0622)
Log total scholarships and fellowships	0.0584 (0.0160)***	0.0732 (0.0189)***	0.0993 (0.0342)***
Log total Pell Grant dollars	0.2286 (0.0416)***	0.2419 (0.0393)***	0.2881 (0.0719)***
Observations	7,075	7,188	3,831
R ²	.9825	.9674	.9301

Note: FTFT = first-time, full-time. Values in parentheses are robust standard errors. All models also include institution and year effects.

Significant at 5%. *Significant at 1%.

tuition and fees at 4-year public institutions was slightly under \$4,200 (measured in 2006 dollars). Yet there is large variation in public university tuition levels, ranging mostly between \$1,400 and \$11,000. Not surprisingly, the cost of tuition at public 4-year colleges and universities is between those at community colleges (\$2,000) and 4-year private universities (\$16,500). Four-year public institutions of higher education also vary widely in their enrollment numbers over the course of the panel, with a mean 12-month unduplicated headcount of 10,700 (ranging from 400 to over 40,000 students).

Multivariate Results

To more fully consider the relationship between tuition increases and enrollment, we present results from our first set of empirical models in Table 2. The columns present the results of our basic specification on each of the three measures of enrollment described above. In columns 1, 2, and 3, respectively, we present estimates of the effect of tuition on total enrollment, total credit hours, and enrollment of FTFT freshmen. All of these models include controls for state characteristics, institution fixed effects, and year effects.

We estimate that the average tuition and fee elasticity of total headcount is −0.0958. Evaluated at the means (approximately \$4,200 tuition and enrollment of 10,700), a \$100 increase in tuition and fees would lead to a decline in enrollment of approximately 25 students, or a little more than 0.23%. This estimate is quite similar to that of Kane (1995), who used state-level data for the 1980s and early 1990s and estimated that a \$100 increase in 1991 dollars resulted in an enrollment decline of just under 0.5%. In comparable dollars, we estimate a decline of just under 0.35%.

The tuition elasticity of credit hours is similar (Table 2, column 2). The enrollment response of freshmen is a bit larger, but less precisely estimated (column 3). This is expected, because students who have not yet matriculated may be most able to change enrollment decisions in response to price changes. But the difference between the tuition elasticities of total headcount and freshman enrollment is not significant at conventional levels.

As expected, enrollment is positively related to the amount of financial aid and grant money available to students. The effects of scholarships and fellowships on enrollment are nearly as large,

and Pell Grant receipts are larger in absolute value than the effects of tuition costs on enrollment. This provides some evidence that aid can be used to offset the impact of increased tuition costs on enrollment. How to interpret these coefficients is made difficult by at least two complications, however. First, scholarships or fellowships and Pell Grants are received by different groups of students, which are likely differently price sensitive. Second, although tuition is largely determined at state or system levels, institutions have some control over the amount of scholarship money available to students and can encourage or enable application for Pell Grants on the part of applicants. Indeed, student aid can be used as a recruiting device. Hence, the coefficient on aid or grant money may be overstated if other factors are associated with the availability or use of student aid at an institution, such as academic counseling or student support services.

The cross-price elasticities of private college tuition and community college tuition are of the expected signs but are statistically insignificant. As before, it appears that freshmen may be more responsive to price changes at the margin. Yet these enrollment effects are small and always insignificant.

Results by Institution Type

We next turn to the question of whether the enrollment response of tuition changes is different for different types of public universities and colleges. In Table 3, we present results of models identical to those in Table 2 but estimated separately for Research I, Research II, comprehensive, and liberal arts institutions. A striking pattern emerges in comparing enrollment responses to tuition across institution types. The tuition elasticity of enrollment is largest at Research I universities. Enrollment also falls with tuition at Research II universities, but the elasticity of total headcount is about two thirds as large as that at Research I schools. If we look at credit hours, tuition elasticities are quite similar. But if the measure is enrollment of FTFT freshmen, enrollment appears much more price responsive at Research I schools (even if less precisely estimated). For neither comprehensive nor public liberal arts colleges are there significant negative enrollment effects of tuition increases.

Important for understanding the relative price sensitivity of enrollment at these different types of institutions, the average amount of aid available to students is mostly strongly related to enrollment at comprehensive and liberal arts institutions. Although Pell Grant dollars are associated with higher enrollments even at research-intensive colleges, the magnitudes of this aid effect are smaller than at comprehensive, liberal arts, and even Research II schools. How can a higher price sensitivity be reconciled with lower aid sensitivity at Research I institutions? Two factors seem relevant here. First, one clue lies in substantial intrastate correlation in tuition prices described earlier. Tuition increases at Research I institutions are made alongside tuition increases at comprehensive universities. A second relevant consideration is that these institutions serve different markets. Public Research I institutions are often state flagship schools or universities with national reputations. These institutions compete with other public flagships and private universities and colleges. Research II and comprehensives are typically less selective and may serve as substitutes for more price sensitive students within the state.

One way to examine this hypothesis for the relatively elastic price response of Research I institutions is to focus on the subset that most clearly have national reputations. One common source students used to compare and assess the prestige of various institutions is the annual "Best Colleges" rankings by *U.S. News & World Report*. We use these rankings to identify public colleges and universities in the top 120 institutions in the nation.¹² We then reestimate the models used to generate the results in Table 3 but split the sample into two different groups: those in the top 120 of *U.S. News & World Report's* rankings and all other institutions.

Because top schools compete nationally for students, raising prices may come with more risk than schools whose competitors are other institutions in the same state or system, where relative tuition costs increase at about the same rate. If so, we would expect to see greater price sensitivity at top 120 schools. We present these results in Table 4. In the first panel, we present estimates for top 120 institutions. In the second panel, we present results for all others. We also present the mean percentage of out-of-state FTFT students enrolling at each type of institution.

TABLE 3
Enrollment Effects of Tuition Increases by Institution Type

Variable	Research I			Research II			Comprehensive			Liberal arts		
	Log headcount	Log credit hours	Log FTFT	Log headcount	Log credit hours	Log FTFT	Log headcount	Log credit hours	Log FTFT	Log headcount	Log credit hours	Log FTFT
Log tuition and fees (in state)	-0.2142 (0.0663)***	-0.1763 (0.0644)***	-0.2299 (0.1606)	-0.1381 (0.0686)**	-0.1986 (0.1052)*	-0.0365 (0.1860)	-0.0582 (0.0581)	-0.0290 (0.0634)	-0.0910 (0.1706)	-0.0193 (0.0997)	-0.0143 (0.0969)	0.0170 (0.3138)
Log average private 4-year tuition and fees	0.0976 (0.0379)**	0.0149 (0.0445)	-0.0268 (0.1143)	-0.0274 (0.0809)	0.0518 (0.1187)	0.0476 (0.1848)	0.0335 (0.0469)	0.0626 (0.0525)	0.2809 (0.2182)	-0.0147 (0.1028)	0.1117 (0.0812)	0.0861 (0.1346)
Log average community college tuition and fees	0.0333 (0.0463)	0.0164 (0.0537)	-0.0402 (0.1393)	-0.0327 (0.0660)	0.0013 (0.0817)	-0.0379 (0.1691)	0.0572 (0.0414)	-0.0022 (0.0555)	0.0530 (0.0979)	-0.0919 (0.0852)	-0.0925 (0.0756)	0.2534 (0.2398)
Log average per capita income	-0.1512 (0.3647)	0.0563 (0.3494)	0.1962 (0.3499)	0.7507 (0.4092)*	0.6599 (0.4639)	2.2832 (1.6533)	-0.0936 (0.1904)	-0.0265 (0.2748)	-0.6655 (0.6418)	1.2516 (0.5812)**	0.1704 (0.4706)	0.4287 (0.9665)
Log number of high school graduates	0.2485 (0.0938)***	0.2219 (0.1125)*	0.1573 (0.1307)	0.1609 (0.1328)	0.28450 (0.1701)*	0.5494 (0.2936)*	0.3783 (0.1193)***	0.4883 (0.1184)***	0.6149 (0.2744)**	0.3205 (0.1734)*	0.4538 (0.1577)***	1.2858 (0.3285)***
Log unemployment rate	-0.0790 (0.0437)*	-0.1125 (0.0568)*	0.0952 (0.0785)	0.0588 (0.0981)	0.1328 (0.1248)	0.1112 (0.1673)	-0.0234 (0.0303)	-0.0469 (0.0357)	0.0093 (0.1061)	0.0899 (0.0778)	-0.0329 (0.0739)	0.0971 (0.1334)
Log total scholarships and fellowships	0.0085 (0.0217)	0.0073 (0.0246)	0.0669 (0.0318)**	0.1097 (0.0453)**	0.1039 (0.0556)*	0.1649 (0.0918)*	0.0395 (0.0233)*	0.0616 (0.0276)**	0.0983 (0.0482)**	0.0476 (0.0518)	0.0791 (0.0549)	0.2190 (0.1106)*
Log total Pell Grant dollars	0.1335 (0.0556)**	0.1145 (0.0669)*	0.1828 (0.0867)**	0.1509 (0.0626)**	0.1665 (0.0649)**	0.4590 (0.2515)*	0.2256 (0.0659)***	0.2404 (0.0622)***	0.2538 (0.1296)*	0.4194 (0.1216)***	0.4109 (0.0895)***	0.3643 (0.1604)**
Observations	1,343	1,358	741	832	848	463	3,532	3,582	1,969	862	887	467
R ²	.8707	.9052	.8572	.9691	.9388	.8259	.9688	.9379	.8704	.9514	.9306	.8829

Note: FTFT = first-time, full-time. Values in parentheses are robust standard errors. All models also include institution and year effects.
*Significant at 10%. **Significant at 5%. ***Significant at 1%.

TABLE 4

Enrollment Impacts of Tuition Increases: Top Public Schools Versus All Others

Variable	Top 120			All others		
	Log headcount	Log credit hours	Log FTFT	Log headcount	Log credit hours	Log FTFT
Log tuition and fees (in state)	-0.2505 (0.0693)***	-0.3051 (0.0903)***	-0.2581 (0.0726)***	-0.0770 (0.0332)**	-0.0844 (0.0368)**	-0.0784 (0.1078)
Log average private 4-year tuition and fees	0.2389 (0.1287)*	0.0544 (0.2306)	0.1935 (0.1245)	0.0001 (0.0339)	0.0345 (0.0447)	0.1284 (0.1181)
Log average community college tuition and fees	0.0504 (0.0393)	0.0452 (0.0616)	0.1279 (0.0631)**	0.0027 (0.0309)	-0.0077 (0.0378)	0.0283 (0.0792)
Log average per capita income	-0.3810 (0.4210)	0.5301 (0.4957)	-0.4077 (0.4712)	0.3770 (0.1700)**	0.1777 (0.1955)	0.0603 (0.5015)
Log number of high school graduates	0.2569 (0.1151)**	0.2271 (0.1856)	0.1512 (0.1400)	0.3081 (0.0719)***	0.3985 (0.0731)***	0.6212 (0.1627)***
Log unemployment rate	-0.0969 (0.0640)	-0.0401 (0.0688)	0.0580 (0.0492)	0.0107 (0.0268)	-0.0198 (0.0318)	0.0519 (0.0697)
Log total scholarships and fellowships	0.0175 (0.0258)	0.0533 (0.0387)	0.0443 (0.0282)	0.0621 (0.0177)***	0.0745 (0.0209)***	0.1068 (0.0376)***
Log total Pell Grant dollars	0.1108 (0.0856)	0.1252 (0.0754)	0.1290 (0.0817)	0.2363 (0.0448)***	0.2525 (0.0427)***	0.2947 (0.0797)***
Observations	768	777	417	6,307	6,411	3,414
R ²	.9071	.9480	.9743	.9829	.9611	.9129
Mean percentage out of state		22.3			14.7	

Note: FTFT = first-time, full-time. Values in parentheses are robust standard errors. All models also include institution and year effects. The difference in mean percentage out-of-state FTFT enrollments between top 120 schools and all others is statistically significant at the 1% level.

*Significant at 10%. **Significant at 5%. ***Significant at 1%.

At the top 120 schools, there is evidence that enrollment is more sensitive to tuition. The price elasticity of 12-month headcount and total credit hours are significant and substantially larger at top schools. Furthermore, enrollment is less sensitive to aid at top 120 schools than at other institutions. These patterns in price and aid sensitivity are consistent with students opting out of top 120 schools for competitors as price rises, while finding a way to pay tuition bills at other state schools at which students may have fewer options. The percentage of out-of-state FTFT freshman enrolling in top 120 schools is also appreciably larger than the corresponding percentage enrolling in other public 4-year institutions.¹³

Another explanation for larger enrollment responses at Research I institutions is due to the greater composition of out-of-state students in their total enrollment. Because in-state and out-of-state tuition follow similar trends, if out-of-state students are more price sensitive

because they have alternatives in their state of residence, we would expect higher overall enrollment response at Research I institutions. To examine this possibility, we first graph (Figure 5) the increases in nominal out-of-state and in-state tuition costs separately, comparing both with the CPI for all urban consumers, for each type of public 4-year institution. Across all types of schools, out-of-state tuition cost patterns clearly mirror in-state cost trajectories. Therefore, we continue to use changes in real in-state tuition costs as our main independent variable of interest; however, using additional information from IPEDS, we split the FTFT outcome into two separate variables: the total number of out-of-state FTFT students and the total number of in-state FTFT students. We then estimate models otherwise identical to those in columns 3 and 6 of Table 4 on both of these outcomes by type of institution. Table 5 presents these results. Additionally, we present the mean percentage

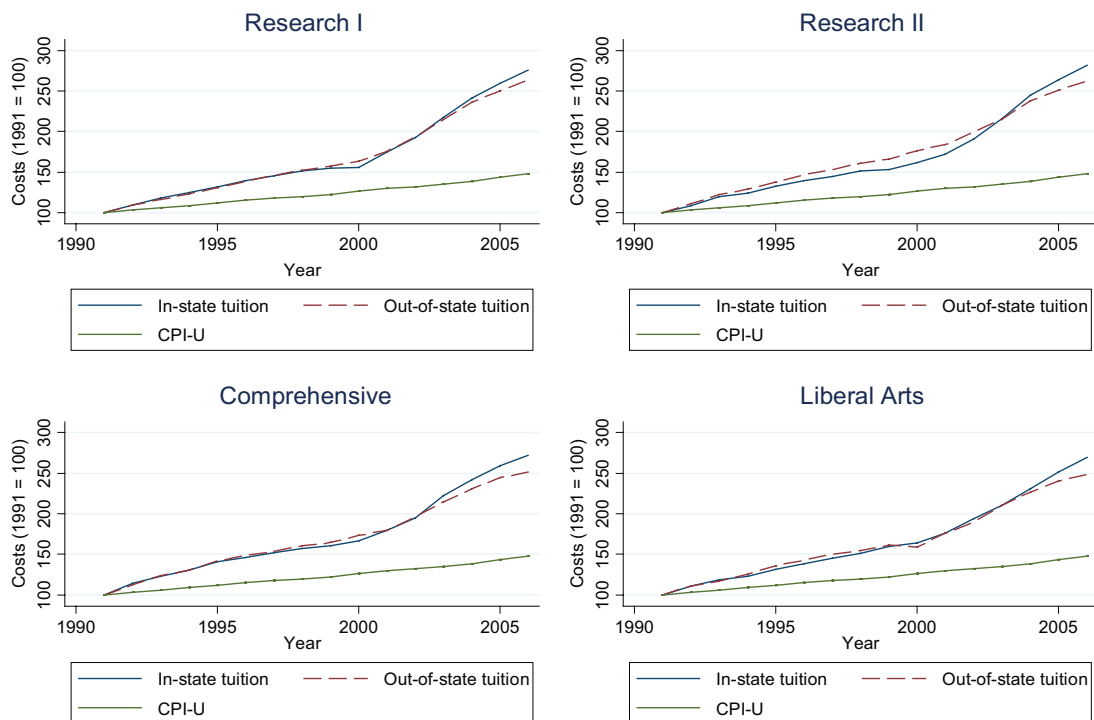


FIGURE 5. *In-state versus out-of-state tuition prices by institution type.*

Note: CPI-U = Consumer Price Index for all urban consumers.

of out-of-state enrollments applicable to each type of institution.

Examining the results in Table 5, we again see a much higher tuition price sensitivity at Research I institutions for both in-state and out-of-state FTFT enrollments, with little sensitivity at other institution types. The reverse pattern holds for enrollment sensitivity to financial aid. The enrollment sensitivity of out-of-state students to tuition increases appears to be slightly greater than the price sensitivity of in-state students, but this difference is not statistically significant. We also see that incoming FTFT freshman cohorts at Research I schools are about 19.4% out-of-state students, compared with 15.1% at Research II and 12.9% at comprehensive institutions. Both of these differences in mean percentage of out-of-state FTFT students are statistically significant at conventional levels.

The evidence from Tables 3, 4, and 5 of higher price sensitivity but lower aid sensitivity at top 120 and Research I institutions raises general questions about enrollment patterns at public 4-year colleges and universities, beyond the

implications of tuition on enrollment at single institutions. One implication may be a shift of students from higher income families to private institutions or public universities in other states, along with a shift of students from lower income families to less expensive public universities within the state.¹⁴ This would suggest a redistribution of students across public colleges and universities within a state, with those most financially able leaving the system, and others scaling back to enroll at more affordable institutions. Obviously, student-level data are needed to more fully test these hypotheses.

Effects of Large Tuition Increases

Another way in which the average price response estimated in Table 2 may not be fully informative is the possibility that the average conceals relatively large enrollment declines during years following especially large real tuition increases. To explore this possibility, we first consider patterns of enrollment before and after large tuition increases.

TABLE 5
In-State Versus Out-of-State Enrollment Effects of Tuition Increases

Variable	Log FTFT							
	Research I		Research II		Comprehensive		Liberal Arts	
	Out of state	In state	Out of state	In state	Out of state	In state	Out of state	In state
Log tuition and fees (in state)	-0.4376 (0.1874)**	-0.3366 (0.1362)**	-0.1481 (0.2955)	-0.0110 (0.1999)	-0.2448 (0.1786)	-0.2022 (0.1490)	-0.1414 (0.3081)	-0.2835 (0.1800)
Log average private 4-year tuition and fees	0.0725 (0.1616)	0.0705 (0.0965)	0.0012 (0.2165)	0.0147 (0.1873)	0.1685 (0.1340)	0.3767 (0.2419)	0.1772 (0.2331)	0.0373 (0.1367)
Log average community college tuition and fees	-0.0773 (0.1670)	0.0359 (0.1005)	-0.0559 (0.2671)	0.0487 (0.1820)	-0.0368 (0.1479)	0.0191 (0.0949)	0.2412 (0.2305)	0.1746 (0.1418)
Log average per capita income	0.3099 (0.8252)	-0.3293 (0.5471)	1.7194 (1.9642)	2.6293 (1.7417)	-1.3056 (0.6814)*	-0.5603 (0.7259)	0.6338 (1.4944)	-0.0412 (0.9368)
Log number of high school graduates	-0.4052 (0.3932)	0.2392 (0.1325)*	0.0784 (0.5038)	0.7405 (0.3026)**	-0.0235 (0.3438)	0.6384 (0.2898)**	0.1388 (0.6486)	1.6916 (0.3013)***
Log unemployment rate	0.0803 (0.1532)	0.0571 (0.1031)	-0.0455 (0.2130)	0.1229 (0.1747)	-0.2311 (0.1173)**	0.0708 (0.1274)	0.0533 (0.2011)	-0.0416 (0.0864)
Log total scholarships and fellowships	0.1139 (0.1210)	0.0122 (0.0839)	0.0769 (0.1400)	0.2123 (0.0995)**	0.1043 (0.0539)*	0.1062 (0.0494)**	0.0713 (0.2141)	0.1936 (0.1062)*
Log total Pell Grant dollars	0.1379 (0.1107)	0.1483 (0.0755)*	0.5229 (0.2756)*	0.4804 (0.2419)*	0.1842 (0.0943)*	0.2678 (0.1348)**	0.3087 (0.3002)	0.1994 (0.1390)
Observations	739	740	463	462	1,954	1,959	459	465
R ²	.9370	.8387	.8821	.8761	.9180	.9133	.8893	.9292
Mean percentage out of state	19.4		15.1		12.9		19.4	

Note: FTFT = first-time, full-time. Values in parentheses are robust standard errors. All models also include institution and year effects. The differences in mean percentage out-of-state FTFT enrollments between Research I and Research II and between Research I and comprehensive institutions are both statistically significant at the 1% level.

*Significant at 10%. **Significant at 5%. ***Significant at 1%.

In Figure 6, we present time series of enrollments, net of institution fixed effects, at institutions that raised tuition from one year to the next by more than 15% compared with institutions that did not. To set up a simple difference-in-differences style comparison, we center the series for institutions with large hikes around the year of the first hike large hike (Year 0). To develop the counterfactual, we center the time series for institutions without large hikes in 1998, which was the mean year in which these large hikes were implemented. There are interesting and suggestive differences in the patterns of enrollment in institutions making large tuition hikes compared with those that do not. Enrollment falls at institutions making

large hikes after the first, second, and third years following large tuition hikes, relative to those without such large tuition increases. Interestingly, enrollments grow a bit faster in the years preceding a large tuition increase than they do at comparable institutions that do not adopt large tuition increases. This raises the possibility that large changes in tuition are more likely at institutions experiencing especially large growth. This is a topic we return to in the following discussion.

To more fully investigate the presence of disproportionate enrollment effects of particularly large tuition increases, we present results in Table 6 from estimations using the setup described by Equation 2. In the first set of

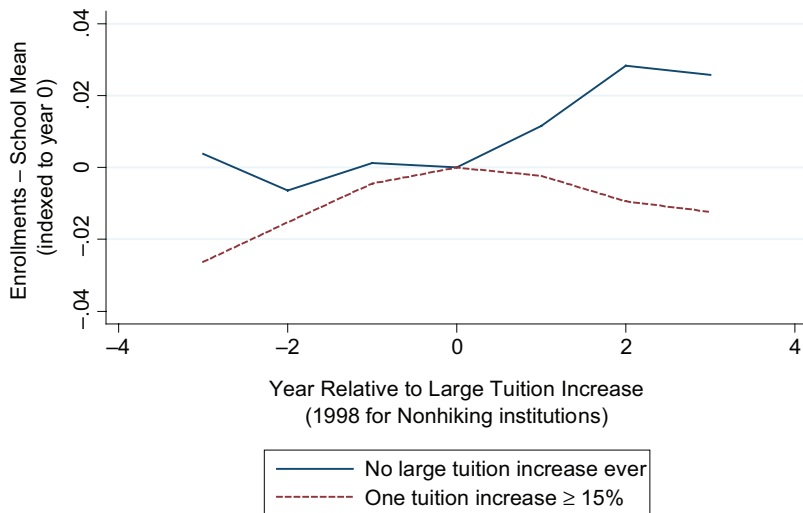


FIGURE 6. *Enrollment patterns by presence of large tuition increases.*
 Note: Institutions that adopted back-to-back hikes of 15% or greater are not included.

columns, we present results from models in which we augment the specification in Table 2 to include an indicator denoting whether the current-year tuition costs were at least 10% higher than the prior year, along with a series of lags. In the next three columns, we present results for the same models in which the threshold for what determines a large hike increases from a 10% increase to at least a 15% increase in real tuition.

The interpretation of the elasticities in the first row of Table 2 remains the same as in previous models. Regardless of whether we define especially large tuition increases to be above 10% or above 15%, the total enrollment falls by about -0.09% for each percentage increase in tuition. Recall that the estimate from Table 2 comparable with the results in columns 1 and 4 here was -0.0958 . If enrollment fell markedly after large tuition increases, we would see negative coefficients on the lag variables that pick up changes in enrollment above institution trends. Furthermore, one might expect the absolute value of these lag-year shifters to increase, as students are better able to adjust. However, we see limited evidence of such nonlinear effects.

If there are any additional declines in enrollments following very large tuition increases, above and beyond the linear impacts, we see these declines in the third post-hike year and only for total credit hours. If we focus on tuition increases of 10% or more, total credit hours are

expected to fall an additional 1.3% the 3rd year after the hike was instituted. Similarly, when a large hike is defined as an increase in tuition costs of 15% or more, we predict total credit hours 3 years out to fall off by an additional 1.9%. The enrollment coefficients for FTFT students border on significance. For example, the t statistics corresponding to the coefficients on the 2nd and 3rd post-hike years in column 3 are both about 1.4. Taken together, these findings suggest that although students may not drop out of college in response to very large tuition increases, they may reduce course loads or postpone entering college. Finally, corroborating the run-up in prehike enrollments we observe in Figure 6, the coefficients on the contemporaneous tuition hike indicators in columns 2 and 4 of Table 6 suggest that enrollments were indeed rising a little faster than normal at institutions that later chose to adopt especially large tuition increases.

Discussion

Although the patterns of intertemporal tuition changes across public 4-year institutions within states along with the inverse relationship between tuition increases and state appropriation levels provide confidence that the tuition increases observed during the course of this panel can be taken as exogenous at the institution level, other concerns related to interpreting

TABLE 6

Enrollment Effects of Large Tuition Hikes: Full Sample

Variable	10%			15%		
	Log headcount	Log credit hours	Log FTFT	Log headcount	Log credit hours	Log FTFT
Log tuition and fees (in state)	-0.0882 (0.0336)***	-0.0797 (0.0500)	-0.0349 (0.1131)	-0.0847 (0.0341)**	-0.0813 (0.0487)*	-0.1059 (0.1082)
Log average private 4-year tuition and fees	0.0258 (0.0333)	0.0561 (0.0481)	0.1555 (0.1165)	0.0261 (0.0335)	0.0559 (0.0480)	0.1549 (0.1167)
Log average community college tuition and fees	0.0061 (0.0277)	-0.0094 (0.0361)	0.0525 (0.0707)	0.0083 (0.0274)	-0.0078 (0.0354)	0.0391 (0.0687)
Log average per capita income	0.2342 (0.1585)	0.2592 (0.1897)	-0.0559 (0.4593)	0.2443 (0.1581)	0.2649 (0.1891)	-0.0377 (0.4593)
Log number of high school graduates	0.3149 (0.0713)***	0.4041 (0.0757)***	0.5700 (0.1398)***	0.3184 (0.0708)***	0.4113 (0.0754)***	0.5825 (0.1406)***
Log unemployment rate	-0.0017 (0.0254)	-0.0131 (0.0297)	0.0671 (0.0629)	0.0008 (0.0252)	-0.0119 (0.0294)	0.0639 (0.0631)
Log total scholarships and fellowships	0.0431 (0.0149)***	0.0619 (0.0179)***	0.0967 (0.0312)***	0.0429 (0.0148)***	0.0617 (0.0179)***	0.0988 (0.0309)***
Log total Pell Grant dollars	0.2090 (0.0441)***	0.2151 (0.0417)***	0.2717 (0.0810)***	0.2078 (0.0439)***	0.2144 (0.0415)***	0.2761 (0.0819)***
Large hike dummy (year of hike)	0.0091 (0.0063)	0.0205 (0.0124)*	-0.0270 (0.0172)	0.0154 (0.0080)*	0.0306 (0.0214)	-0.0028 (0.0224)
1st year after hike	0.0020 (0.0060)	-0.0013 (0.0076)	-0.0020 (0.0175)	-0.0025 (0.0076)	-0.0039 (0.0109)	0.0156 (0.0217)
2nd year after hike	-0.0076 (0.0073)	-0.0092 (0.0080)	-0.0217 (0.0155)	-0.0157 (0.0104)	-0.0091 (0.0086)	-0.0079 (0.0217)
3rd year after hike	-0.0038 (0.0056)	-0.0129 (0.0075)*	-0.0238 (0.0164)	-0.0092 (0.0067)	-0.0194 (0.0100)*	-0.0145 (0.0283)
Observations	6,399	6,509	3,75	6,399	6,509	3,746
R ²	.9830	.9663	.9300	.9831	.9663	.9299

Note: FTFT = first-time, full-time. Values in parentheses are robust standard errors. All models also include institution and year effects.

*Significant at 10%. **Significant at 5%. ***Significant at 1%.

the enrollment-tuition relationships estimated here as evidence of demand elasticity may remain.

An important assumption underlying all our models is that there has been no sharp increase in the supply of students within states, above prevailing trends that are associated with tuition changes. In addition to the intertemporal within-state tuition cost patterns described above, we also examined the enrollment effects of tuition increases for public in-state students only. Out-of-state tuition increases mirrored in-state increases and patterns of enrollment sensitivity across different types of 4-year public institutions did not meaningfully differ for in-state versus out-of-state student populations.

An additional concern about interpreting our main estimates as demand elasticities is the possibility that schools might alter admissions criteria at the margin as a means to offset the negative enrollment impacts of tuition increases. If public 4-year institutions lower their admissions standards, or simply admit more students following large tuition increases, these behaviors could affect overall enrollment numbers as well as the impact of large tuition increases on future enrollments. One way to assess the importance of this concern is to examine the year-to-year change in the percentage of students admitted to see if it varies systematically with year-to-year changes in tuition costs. After all, admissions

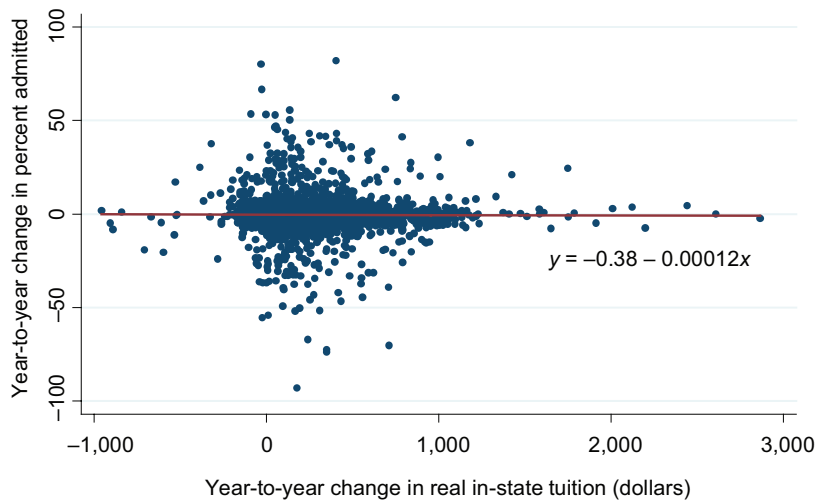


FIGURE 7. *Yearly tuition increases versus admission rates: 4-year public institutions, 2001 to 2007.*
 Note: The line is a linear fit of change in admission percentage on change in real tuition.

decisions are the only real lever institutions have to manage enrollment. If institutions are planning enrollment in response to higher levels of tuition and fees, we would expect higher rates of admission as a means to fill seats that might be vacated or because of lower anticipated yield at the higher level of tuition.

In Figure 7, we summarize the relationship between admission decisions and recent tuition changes for institutions in our panel.¹⁵ The y -axis measures year-to-year changes in the percentage of applicants offered admission, while the x -axis measures year-to-year changes in tuition and fee prices. Over the scatterplot is a linear fit of the relationship, confirming that there is no evidence in our panel that rates of admission increase with tuition.¹⁶ Hence, we find no evidence that public 4-year colleges and universities were engaged in strategic behaviors to increase enrollment in response to tuition changes during the later portion of our panel, a period rife with large tuition increases. Indeed, it is not clear that they could do much about it. Because of the vagaries of yield, knowing how many students to admit to meet targets as tuition prices change is surely difficult. Furthermore, there is nothing admissions officers can do about lower retention rates that are associated with higher tuition, and increases in drop-out at that margin could swamp any strategic efforts admissions officers do make.

Although none of these checks is ironclad, together they suggest our main identifying assump-

tions hold. To the extent that universities and colleges were responding in other ways to tuition changes that might also affect enrollment numbers, such behaviors would cause us to underestimate the true demand elasticity of enrollment. Therefore, at a minimum, our estimates reliably identify a lower bound of the true demand sensitivity of enrollment to tuition changes.

Conclusion

During the past decade, there have been exceptional and perhaps unprecedented increases in tuition at public colleges and universities. Poor economic conditions and subsequent state budget cuts have created a fertile landscape for large tuition increases. Those pressures have not abated. We survey the terrain of public higher education between 1991–1992 and 2006–2007 to update what is known about the relationship between tuition and enrollment. We make use of the variation in the timing and magnitude of sometimes very large tuition increases to examine patterns of enrollment.

An important empirical finding to derive from our work is that despite increases in the rate of real tuition growth, there is no evidence that the tuition elasticity of enrollment at public 4-year institutions has increased. We estimate that the average tuition and fee elasticity of total headcount is -0.0958 . So, at the mean, a \$100 increase in tuition and fees (in 2006 dollars) would lead to a decline in enrollment of a little

less than 0.25%. This is quite similar to estimates of tuition elasticities from the 1980s and early 1990s (Kane, 1995; Heller, 1996).

Our estimates suggest that tuition can be used as a lever to offset revenue losses from declining appropriations. At the means of enrollment and tuition price, our results imply that a 5% increase in tuition (about \$210) would result in an enrollment decline of about 51 students and the loss of about \$225,000 in tuition from these students. But the higher price charged to remaining students would bring in an additional \$2.24 million in tuition revenue. So, if net revenue in the short run is the only concern for an institution, tuition is clearly a mechanism for augmenting revenue. Obviously, there may be political or other considerations important for public institutions that complicate this calculus.

We find limited evidence that unusually large year-to-year tuition increases (e.g., real increases in excess of 15%) have disproportionately large impacts on enrollment. If they do, such impacts appear to manifest as less drastic decisions on the part of students, such as decreasing future course loads (in terms of the number of credit hours taken).

We also find substantial differences in enrollment responses at different types of colleges and universities. We find larger effects of tuition increases on enrollment at Research I and top 120 public universities than we do at comprehensive universities and public liberal arts colleges. Moreover, enrollment is less sensitive to aid at Research I universities and those in the top 120 of the *U.S. News & World Report* rankings. At public colleges and universities of this type, it appears that the near-term consequence of increased tuition is a decline in enrollment. On the other hand, at comprehensive universities, it appears that tuition increases do not necessarily mean lower enrollment; rather, they mean more reliance on aid for the students who do enroll.

Institution-level panel data, like ours, are ultimately quite limited for the purposes of understanding consequences of tuition increases on individual students, educational attainment, and the public system of higher education more broadly. The enrollment responses at the institution level are consistent with softness in demand at the top tier of public colleges and universities. This could arise because students at these insti-

tutions view schools in other states, or private universities, as substitutes. It could also arise because these schools are relatively expensive within state systems, and price increases induce some students to substitute to choose less expensive options within the state. Of course, both of these explanations may be relevant for different sets of students. In either case, shifts would be consistent with a cascading effect in the public university system that implements large tuition increases: As tuition increases, the outflow is largest at the most prestigious state universities as students there leave for other states, private colleges, or less expensive options within the state. At the relatively inexpensive but less selective Research II and comprehensive state universities, the loss of students who substitute down to less expensive options such as community colleges could be offset by students who once would have gone to the state flagship but can no longer afford it.

Although we cannot assess the relative merits of explanations such as these using institution-level data, doing so is clearly important. Public institutions enroll the vast majority of students in American higher education, and relative price changes may result in substantial shifts across institutions, or enrollment intensity within institutions. Shifts in where students enroll and the number of credit hours taken may play a role in helping us understand why college completion rates have declined, even as enrollment rates have risen. Bound, Lovenheim, and Turner (2010) examined initial enrollment decisions of students from the 1970s and 1990s and find important shifts toward less selective postsecondary institutions, where student support and completion rates are lower. Furthermore, they found that “student observables explain virtually none of the observed cross-cohort shifts in initial school choice” (p. 142). One possibility suggested by our findings is that some of these shifts were due to relative price changes.

Another policy question to which the current findings are related is a concern about increasing stratification of student and institutional quality. An important dimension of this has been at the public-private margin: Kane, Orszag, and Gunter (2003) reported that the combination of falling state appropriations and political constraints on tuition increases have led to relative

Appendix

Variable Sources and Availability

Variable	Availability (Academic Years)	Source	IPEDS Variable Names, Table, or Specific Survey Element
Costs			
In-state tuition and fees	1990–1991 to 2007–2008	IPEDS	Tuition and fees, full-time undergraduate, in-state and published in-state tuition and fees
Average 4-year private tuition and fees measure (by state, year)	1993–1994 to 2006–2007	<i>Digest of Education Statistics</i> (NCES), multiple years	Postsecondary tables, by year and state (for recent years) and State Comparisons of Education Statistics (1969–1997)
Average community college tuition and fees (by state, year)	1993–1994 to 2006–2007	<i>Digest of Education Statistics</i> (NCES), multiple years	Postsecondary tables, by year and state (for recent years) and State Comparisons of Education Statistics (1969–1997)
Outcomes			
12-month unduplicated undergraduate headcount	1991–1992 to 2006–2007	IPEDS	12-month unduplicated undergraduate headcount
Total undergraduate credit hours (measure of instructional activity)	1991–1992 to 2006– 2007 (linearly interpolated: 1994– 1995 and 1996–1997)	IPEDS	12-month instructional activity credit hours: undergraduates
FTFT undergraduate enrollment	1999–2000 to 2006–2007	IPEDS	Number of FTFT students in cohort who are in-state/out-of-state
Other controls			
State unemployment rates (by year)	1991–1992 to 2007–2008	BLS	Local Area Unemployment Statistics (multiscreen option) ^a
Total personal income (by state, year)	1990–1991 to 2006–2007	BEA	Table 658: “Personal Income in Current and Constant (2000) Dollars by State: 1979-2006”
State populations (by year)	1990–1991 to 2007–2008	Census	National and State Population Estimates: 2007 vintage for later years, 1990s archive for earlier years
Average per capita income (by state, year)	1990–1991 to 2006–2007	Created	
Public high school graduates (by state, year)	1990–1991 to 2005– 2006 (linearly interpolated: 1991– 1992 to 1994–1995, 1996–1997, and 1998–1999)	<i>Digest of Education Statistics</i> (NCES), 2007	Table 101: “Public High School Graduates, by State or Jurisdiction: Selected Years, 1980-81 Through 2005-06”
Total Pell Grants	1990–1991 to 2006–2007	IPEDS	(Total) Pell Grants
Total scholarships and fellowships	1990–1991 to 2006–2007	IPEDS	Total (gross) scholarships and fellowships

Note: BEA = U.S. Bureau of Economic Analysis; BLS = U.S. Bureau of Labor Statistics; FTFT = first-time, full-time; IPEDS = Integrated Postsecondary Education Data System; NCES = National Center for Education Statistics.

a. <http://www.bls.gov/data/#unemployment>.

declines in per student spending at public colleges and universities. The ratio declined from about 70% public to private per student spending in the mid-1970s to 58% by the mid-1990s. More generally, Hoxby (2009) found that student quality, tuition, and subsidies have stratified markedly across institutions ranked by quality. Our finding that large tuition increases lead to enrollment declines at selective public universities but steady enrollment, and perhaps larger debt, at less selective colleges suggests there may be a link between policies that offset general revenue declines with tuition increases and this larger landscape of stratification.

Notes

1. Nonetheless, the correlation coefficient on intertemporal variation in tuition at these institutions is .984.
2. It is important to note that enrollment responses to changes in tuition price need not be equivalent to responses to identical changes in the availability of financial aid, because the latter require more information and effort to adequately price.
3. The Bureau of Labor Statistics calculates and publishes a variety of CPI measures. We use the multi-screen database tool to extract yearly CPI values for all urban consumers (<http://www.bls.gov/data/#prices>).
4. This figure is the average R^2 value across regressions by state of real tuition costs on institution effects.
5. Some basic attempts at modeling demand and supply simultaneously can be found for specific types of institutions, such as doctorate-granting institutions (Koshal & Koshal, 1994), private liberal arts colleges (Koshal & Koshal, 1999), and public comprehensive universities (Koshal & Koshal, 1998). Yet these studies all used data from just 1 academic year, and they suffer from the common inability to find credible factors that shift either supply or demand without affecting the other.
6. Certainly students might adjust the number of classes (or credits) differently at the margin on the basis of whether institutions price by credit hour or at a flat semester rate. Yet pricing policies differ by institution, ranges of credit hours, and by whether a student is full- or part-time. It is reasonable to argue that this adjustment at the margin in response to tuition changes would be greatest at institutions that price by credit hour. We capture the average effect across all institutions.
7. We deflate all monetary values to 2006 dollars using the CPI for all urban consumers.
8. This figure includes grants, loans, work study, and "other" federal and nonfederal aid.

9. Even in the presence of financial aid to help reduce costs, students are likely to respond first and most strongly to the "sticker price" of attendance. For instance, decisions about some types of financial aid come much too close to the time students must make actual attendance decisions. Furthermore, many students do not even apply for financial aid, because of the complicated nature of the forms and process (Dynarski & Scott-Clayton, 2006).

10. We also estimated models in which we clustered on state, because of the substantial relationship between intertemporal patterns of tuition across institutions within states. Standard errors from models clustered on states were indistinguishable from models clustered on institutions.

11. There is evidence that families do not substantially adjust college savings plans in response to changes in financial aid incentives (Long, 2004; Monks, 2004).

12. We identified schools in our sample as being top 120 schools by whether they appeared on the "Best Colleges 2000" list published in the August 1999 issue of *U.S. News & World Report* (which is approximately the midpoint of our panel).

13. This difference in mean out-of-state FTFT enrollments is statistically significant at conventional levels.

14. In our sample, the average cost of enrollment at a Research I university is \$4,837, compared with \$4,390 at Research II schools and \$3,869 at comprehensive institutions.

15. We focus on the latter half of our panel, because admissions data are available from IPEDS only for 2001 through 2007.

16. This conclusion is robust to whether we use contemporaneous or lagged changes in year-to-year tuition costs and to whether we use changes in the percentage admitted or changes in the percentage enrollment yield (defined as the number of students enrolled divided by the number admitted) as the outcome of interest.

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